

Ramgopal Reddy

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EDUCATION

Northeastern University

Boston, MA

Master of Science in Data Analytics Engineering

May 2027

Relevant Coursework: Foundations of Data Analytics, Database Management, Data Mining, Machine Learning Operations, Deep Learning for AI, Applied Generative AI

The LNM Institute of Information Technology

Jaipur, India

Bachelor of Technology in Computer Science

May 2025

Relevant Coursework: Algorithms, Data Structures, Object-Oriented Programming, Operating Systems, Databases, Computer Networks, Artificial Intelligence, Machine learning, Deep Learning, Reinforcement Learning

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, C++, HTML/CSS

Libraries: Pandas, NumPy, Matplotlib, Sci-kit Learn, Transformers, NLTK

Frameworks: Pytorch, React, Flask

Developer Tools: Git, Jupyter, Microsoft PowerBI, Excel

Statistics: Probability Distributions, Hypothesis Testing, Regression Analysis, ANOVA, Time Series Analysis

PROJECTS

Volatility Forecasting & Retirement-Outcome Engine

September 2026

- Designed and built an end-to-end MLOps system that forecasts 21-trading-day-ahead realized volatility for a 30-name, 10-sector equity universe and operationalizes the full lifecycle
- Engineered a leakage-safe data pipeline over 138K+ rows, benchmarking a gradient-boosted-trees challenger against GARCH and HAR-RV econometric baselines under walk-forward CV with a 21-day purge and 5-day embargo
- Automated a weekly challenger retraining loop gating every promotion on a 7-criterion rule with logged, reversible decisions, plus a monitoring loop on 21-day-delayed labels feeding a 10,000-path Monte Carlo retirement simulator

Auxless - Music Recommender Engine

April 2026

- Architected a hybrid recommendation engine combining Content-Based Filtering (CBF), Collaborative Filtering (CF), and a PyTorch GRU sequential model over an 81,000-song BigQuery catalog, encoding tracks as 386-dimensional sentence-transformer embeddings
- Implemented two-level bias detection and mitigation across genre and country slices — upsampling slices below 5% and capping dominance above 60% — producing auditable before/after mitigation reports for every run.
- Automated an end-to-end CT/CM pipeline using KL-divergence drift detection to gate GRU retraining behind a 1% validation-improvement deploy check, scheduled every 3 days via Cloud Scheduler with Cloud Monitoring alerting

Chess Engine

Jan 2025

- Engineered a python based chess engine from scratch, implementing move generation, game state tracking and advanced mechanics such as castling and En Passant ensuring effective state management
- Enhanced engine intelligence by integrating minimax and alpha-beta pruning algorithms, reducing search space over 90% and enabling 3x faster decision making, improving AI efficiency

EXPERIENCE

Summer Research Intern

May 2023 – July 2023

LUSIP - 2023 (User Level Intervention in Hate Speech Systems)

- Analysed over 4000+ social media posts and annotated a functional set of 6 categories for Hate and 8 for Counter-hate speech essentially capturing how these users interact with each other.
- Developed a mixture of machine learning and deep learning models that incorporate the context of parent tweet in-order to better detect and classify Counter-hate speech, thereby achieving a total accuracy of 72%.

ACHIEVEMENTS

- Core Council Member and Treasurer, LNMIIT ACM Student Chapter (2024–2025): Led a 20-member team to organize numerous research-driven technical events and workshops, fostering a strong research culture on campus